

Tanmai Kumar Ghosh

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Professional Experience

Bangladesh Japan IT Limited

Dec 2021 - Jul 2023

Software Engineer

Dhaka, Bangladesh

- Position: Software Engineer
- Projects: Creww, Pocketalk
- Developed and maintained RESTful APIs and backend services using Ruby, improving performance through systematic debugging and profiling.
- Developed frontend modules with React and TypeScript, implementing production features and ensuring responsive design.
- Integrated automated testing with Test Driven Development (TDD), CI/CD pipelines, and deployment monitoring to improve release quality and reliability.
- Collaborated cross-functionally with product managers, QA, and infrastructure teams in agile delivery cycles, using Git for version control and project management tools.

Misfit Technologies Limited

Oct 2020 - Dec 2021

Junior Software Engineer

Dhaka, Bangladesh

- Position: Junior Software Engineer
- Projects: Queue Management System (Consulate General of Bangladesh, Sydney), Shopoth E-Commerce
- Built and maintained backend services and RESTful APIs using Ruby and Python, adhering to coding standards and optimizing database interactions.

WellDev Bangladesh Limited

Nov 2019 - Sep 2020

Junior Software Engineer

Dhaka, Bangladesh

- Position: Junior Software Engineer
- Project: Welltravel
- Developed and maintained backend APIs using Ruby for a travel booking platform (Welltravel).
- Peer-reviewed the Pull Requests (PR) across the development team.

Skills

- **Programming:** Python, Ruby, Java, JavaScript, TypeScript, C++
- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn
- **Data:** SQL (PostgreSQL, MySQL), NoSQL
- **Web & UI:** Ruby on Rails, Django, ReactJS, HTML, CSS
- **Cloud & DevOps:** AWS, S3, Docker, ElasticSearch, CI/CD (CircleCI)
- **Practices:** Git, Unix, Agile, Object-Oriented programming, Test Driven Development (TDD), Debugging

Education

Boise State University

Aug 2023 - Present

PhD, CS

Boise, ID, USA

- **GPA:** 3.893
- **Coursework:** Design and Analysis of Algorithms, Machine Learning, Artificial Intelligence, Data Science, Advanced Software Engineering, Operating Systems, Advanced Databases, Large Scale Data Analysis

Khulna University

Jan 2015 - Jan 2019

Bachelor's, CS

Khulna, Bangladesh

- **GPA:** 3.40
- **Coursework:** Structured Programming Languages, Object Oriented Programming, Data Structures

Selected Projects

Error-Resilient DNA Data Decoder | [github](#)

- Built a Python-based decoding engine using heuristic search and statistical validation, achieving over 50% improvement in effective storage density under experimental noise conditions.

Interactive Parity Grid Analyzer | [github](#)

- Developed a React + Python visualization tool to inspect decoding behavior, error propagation, and parity relations for rapid experimentation and debugging.